Complex Variables Workshop

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Abstract

This document contains what I intend to cover in the preparatory workshop for the NYU Courant January 2026 complex variables written exams. An emphasis is placed on using key ideas and theorems to solve problems. Thus, most proofs of the main theorems are omitted, unless the proof is particularly enlightening or may appear on the exam. The exact topics covered on each day of the workshop may change depending on how fast/slow we can work through the material.

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Day 1: Review & Differentiation

1.1 Notation and Review

We will define the *complex plane* by

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\} \cong \mathbb{R}^2,$$

where $i^2 = -1$. We write $z \in \mathbb{C}$ to mean that there exists $x, y \in \mathbb{R}$ such that $z = x + iy$. We write $x = \mathrm{Re}z$ and $y = \mathrm{Im}z$, saying that x and y are the *real and imaginary parts* of z, respectively. Addition, multiplication, and division are defined as one would expect,

$$\begin{aligned}(x_1 + iy_1) + (x_2 + iy_2) &= (x_1 + x_2) + i(y_1 + y_2)\\ (x_1 + iy_1)(x_2 + iy_2) &= (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)\\ \frac{x_1 + iy_1}{x_2 + iy_2} &= \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2}.\end{aligned}$$

We will denote the *modulus* of a complex number z by the usual Euclidean norm

$$|z| = \sqrt{x^2 + y^2}$$

and *complex conjugation* by

$$\overline{z} = x - iy.$$

It is often convenient to express a complex number in *polar form*, $z = re^{i\theta}$, where $r = |z|$, $\theta = \arg(z)$, and

$$e^{i\theta} = \cos(\theta) + i\sin(\theta).$$

Example 1.1: Find all cube roots of $1 + i$.

Solution. Write $z^3 = 1 + i = \sqrt{2}e^{i(\pi/4+2\pi k)}$, to find that

$$z = 2^{1/6}e^{\pi/12},\ 2^{1/6}e^{3\pi/4},\ 2^{1/6}e^{17\pi/12}.$$

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We define the *principal argument* of z, denoted by $\mathrm{Arg}(z)$, to be the unique value in $(-\pi, \pi]$ such that

$$\arg(z) = \mathrm{Arg}(z) + 2\pi n \quad \forall n \in \mathbb{Z}.$$

While it is true that $\arg(z_1z_2) = \arg(z_1) + \arg(z_2)$, this is not necessarily true for Arg (you should find a counterexample).

Note that the definition of $e^{i\theta}$ immediately gives a definition of the complex exponential function e^z as

$$e^z = e^xe^{iy} = e^x(\cos(y) + i\sin(y)).$$

In these notes, any non-empty, open, connected set D is called a *domain.*

1.2 Analytic Functions

1.2.1 Derivatives of Complex Functions

A function $f\colon S \subseteq \mathbb{C} \to \mathbb{C}$ can be decomposed into real component functions u, v via

$$f(z) = f(x + iy) = u(x, y) + iv(x, y).$$

The notions of limits and continuity are defined as one would on a metric space, so we will move right on to discussing differentiation of complex functions.

The derivative of f at a point $z_0 \in \mathbb{C}$ is defined as

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

Taking $\delta z = z - z_0$, this is equivalent to

$$f'(z_0) = \lim_{\delta z \to 0} \frac{f(z_0 + \delta z) - f(z_0)}{\delta z}.$$

The classical example of a complex function that is not differentiable is the following:

Example 1.2: The conjugation map $z \mapsto \overline{z}$ is not differentiable at the origin.

Solution: Assume instead $f(z) = \overline{z}$ were to be differentiable at $z = 0$. Then the limit of the difference quotient exists on any path. Comparing the paths $(\delta x, 0)$ and $(0, \delta y)$, we find a contradiction.

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We note that linearity, the Leibnitz rule, the quotient rule, and the chain rule are all true for differentiable complex functions.

It is also important to note that complex differentiation is not the same as differentiation in $\mathbb{R}^2$. Namely, differentiation in $\mathbb{R}^2$ is like linearization, while complex differentiation is about angle preservation.

Exercise 1.1: Consider the maps $f(x, y) = x$ and $g(z) = \text{Re}(z)$. Show that f is differentiable as a map from $\mathbb{R}^2 \to \mathbb{R}^2$ but g is not complex differentiable.

1.2.2 Cauchy-Riemann Equations

Perhaps the most important system of partial differential equations (PDEs) in complex analysis are the *Cauchy-Riemann equations*:

Theorem 1.1 (C-R Equations)

Suppose $f = u + iv$ is differentiable at a point $z_0 = z_0 + iy_0$. Then the first order partial derivatives of u, v must exist at (x_0, y_0), and they satisfy the Cauchy-Riemann equations

$$u_x = v_y, \quad u_y = -v_x.$$

Moreover,

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

The idea in deriving the C-R equations is to again look at the limit definition of the derivative along the paths $(\delta x, 0)$ and $(0, \delta y)$.

It is important to note that the converse statement is generally not true without additional assumptions on u and v.

Theorem 1.2

Suppose that $f = u + iv$ is defined in some neighbourhood of a point $z_0 = x_0 + iy_0$, and that u, v are continuously differentiable in the neighbourhood and satisfy the C-R equations at (x_0, y_0). Then $f'(z_0)$ exists, with

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

This result is the first instance where we see the importance of the local existence of the derivative. We will soon come back to this when we discuss analyticity and integration.

We quickly note that, in polar coordinates, the C-R equations become

$$ru_r = u_\theta, \quad u_\theta = -rv_r.$$

1.2.3 Analyticity

We say that f is *holomorphic* (or *analytic*) at a point z_0 if there exists a neighbourhood of z_0 such that f is differentiable at each point in that neighbourhood. If f is defined on an open set, it is called holomorphic on that set if it is differentiable at each point. A function is called *entire* if it is defined and holomorphic on all of $\mathbb{C}$. Given a domain D, the set of holomorphic functions on D is denoted by $\mathcal{H}(D)$.

We note that every holomorphic function on a domain D satisfies the C-R equations, and the previous theorems can also be used to prove a function is holomorphic. Another important fact is that if f is holomorphic in a domain D, then the component functions u and v are *harmonic*:

$$\Delta u, \Delta v = 0, \quad \text{where } \Delta = \partial_{xx} + \partial_{yy}.$$

In fact, more can be said, but we first need the following definition. If u, v are harmonic in D and satisfy the C-R equations, then v is called a *harmonic conjugate* of u.

Theorem 1.3

A function $f = u + iv$ is holomorphic in a domain D if and only if v is a harmonic conjugate of u.

Example 1.3: Let $u(x, y) = x^2 - y^2$. Find a harmonic conjugate of u.

Solution: First, note that it is easy to see that u is harmonic. If v is a harmonic conjugate of u, then

$$v_x = -u_y = 2y, \quad v_y = u_x = 2x.$$

Integrating, we see that $v(x, y) = 2xy$ works.

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Exercise 1.2: Let $u(x, y) = e^x(x\cos(y) - y\sin(y))$.

(a) Show that u is harmonic on $\mathbb{R}^2$.
(b) Find a harmonic function v such that $f = u + iv$ is holomorphic on $\mathbb{C}$.
(c) Express $f(z)$ explicitly as a function of z, where $z = x + iy$.

Example 1.4 (January 2017, Problem 1): Define the function

$$f(z) = \begin{cases} \exp\left(-\frac{1}{z^4}\right) & z \neq 0 \\ 0 & z = 0 \end{cases}$$

1. Show that f satisfies the Cauchy-Riemann equations on $\mathbb{C}$.
2. Is f entire?

Solution:

(a) First note that, away from $z = 0$, f is the composition of holomorphic functions. Hence, f satisfies the C-R equations if $z \neq 0$.

Since we are considering the point $(x, y) = (0, 0)$, it suffices to compute the partial derivatives along $x = 0$ and $y = 0$. Decomposing f as $f = u+iv$, it is easy to see that along the axes we have $v \equiv 0$. Meanwhile, $u(x, 0) = \exp(-1/x^4)$, $u(0, y) = \exp(-1/y^4)$. It is a straightforward computation using L'Hopital to show that $u_x(0, 0) = u_y(0, 0) = 0$.

(b) We show f is not continuous at the origin, which proves f is not holomorphic there. To see this, we compute the limit of f as $z \to 0$ along the line $x = y$. Writing $z = re^{i\pi/4}$, this becomes

$$\lim_{r\to 0} f(re^{i\pi/4}) = \lim_{r\to 0} \exp\left(\frac{1}{r^4}\right) = \infty \neq 0 = f(0).$$

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Example 1.5: Let $f(z) = u + iv$ be a holomorphic function on some domain $D \subseteq \mathbb{C}$. Assume there exists $\Phi \in C^1(\mathbb{R}^2 \to \mathbb{R})$ such that

$$\Phi(u(x,y), v(x,y)) = 0 \quad \forall z = x + iy \in D.$$

Assume further that $\nabla\Phi \neq 0$ on $f(D)$. Show that f is a constant.

Solution: Differentiating $\Phi(u,v) = 0$ with respect to x and y and using the C-R equations, we find that

$$\frac{\partial\Phi}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial\Phi}{\partial v}\frac{\partial v}{\partial x} = 0$$
$$\frac{\partial\Phi}{\partial v}\frac{\partial u}{\partial x} - \frac{\partial\Phi}{\partial u}\frac{\partial v}{\partial x} = 0.$$

This is a system of equations where the determinant of the coefficient matrix is $-(\Phi_u^2 + \Phi_v^2) \neq 0$ on $f(D)$. Thus, this system only has the trivial solution for all $z \in D$, so that

$$f'(z) = u_x + iv_x = 0 \quad \forall z \in D.$$

Hence, f is a constant.

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Example 1.6: Let $f \colon D \to \mathbb{C}$ be holomorphic, D a domain. Suppose there exists a constant $\alpha \in \mathbb{R}$ such that

$$\arg(f'(z)) = \alpha \quad \forall z \in D \backslash \{f'(z) = 0\}.$$

Show that f is of the form $f(z) = a + bz$ for some $a, b \in \mathbb{C}$.

Solution: We will use that derivatives of holomorphic functions are holomorphic (we will discuss this more later). Since f' lives on the ray at the angle α, we can write

$$f'(z) = |f'(z)|e^{i\alpha}.$$

Define $g(z) = e^{-i\alpha}f(z)$. Since f is holomorphic on D, so is g. Moreover, $g'(z) = |f'(z)|$, so g' is a real-valued holomorphic function. In particular, the C-R equations guarantee that g' is a constant. Integrating (which we will also discuss later), we see that $g(z)$ is affine.

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Example 1.7 (September 2013, Problem 2): Let $z = x + iy$, $f = f(z) = u + iv$. Assume D is a domain in $\mathbb{C}$, $f \in C^2(D)$. Denote

$$Df = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix}.$$

Suppose for every $z \in D$ that

$$Df(z)^T Df(z) = \lambda(z) I,$$

where I is the 2×2 identity matrix. Show that either f or $\overline{f}$ is holomorphic in D.

Solution: Since $Df^T Df = \lambda I$, we have that

$$\begin{aligned}&(\partial_x u)^2 + (\partial_y v)^2 = \lambda = (\partial_y u)^2 + (\partial_y v)^2,\\&(\partial_x u)(\partial_y u) + (\partial_x v)(\partial_y v) = 0.\end{aligned}$$

Note that the second condition is saying that the columns of Df are orthogonal, while the first says that they have the same length (and λ is a non-negative function). Therefore, each column is precisely a rotation of the other by $\pi/2$ or $-\pi/2$.

Let f_1 and f_2 denote the first and second column of Df, respectively. If f_2 is a rotation by $\pi/2$ of f_1, then

$$\begin{pmatrix} u_y \\ v_y \end{pmatrix} = \begin{pmatrix} -v_x \\ u_x \end{pmatrix}.$$

Thus, the Cauchy-Riemann equations are satisfied, and f is holomorphic.

If instead f_2 is a rotation by $-\pi/2$ of f_1, a similar argument shows f is antiholomorphic. Therefore, for every point $z \in D$, either f is holomorphic or f is antiholomorphic.

To conclude that only one of these cases can be true on all of D, we need to apply the identity theorem (which we will see later). For now, let us assume the following fact: All zeroes of a non-constant holomorphic function on a connected region are isolated.

Let $J(z) = \det\,(Df)$ denote the determinant of the Jacobian of f. Since $f \in C^2$, we have that $J \in C^1$. If $z_0 \in D$ is such that f is holomorphic, then by the C-R equations,

$$J(z_0) = u_x v_y - u_y v_x = u_x(u_x) - u_y(-u_y) = u_x^2 + u_y^2 = \lambda(z_0) \geq 0.$$

Likewise, if $z_1 \in D$ is such that f is antiholomorphic,

$$J(z_1) = -(u_x^2 + u_y^2) = -\lambda(z_1) \leq 0.$$

Since D is connected, we can apply the intermediate value theorem to conclude that J attains zero at least once. However, if $\lambda(z^*) = 0$, this implies that

$$u_x,\ u_y,\ v_x,\ v_y = 0.$$

Let us deal with this more precisely. Let

$$\begin{aligned}U &= \{z \in D : J(z) > 0\}\\V &= \{z \in D : J(z) < 0\}.\end{aligned}$$

Note that, by the intermediate value theorem, U and V cannot consist of isolated points. Moreover, by continuity, U and V are open. Define also

$$\Gamma = \{z \in D : J(z) = 0\}$$

If Γ contained an open set, then by analytic continuation f would be constant on D, so we can assume this is not the case. Assume U and V are both non-empty. Since D is connected, one cannot pass from U to V without intersecting Γ.

Let $g(z) = f'(z)$ for all $z \in D$. Then g is holomorphic in U, continuous on $\overline{U}$, and vanishes on $\Gamma \cap \partial U$. If Γ has an accumulation point, then the identity theorem would ensure that g vanishes on all of U, so that f is constant. Thus, Γ can only consist of isolated points. But then $D\backslash\Gamma$ would still be connected, contradicting the intermediate value theorem applied to J (since then the image would be the union of the disjoint intervals $(-\infty, 0)$ and $(0, \infty)$!). Hence, either $U = D$ or $V = D$.

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Example 1.8 (September 2024, Problem 5): Suppose $f\colon D \to \mathbb{C}$ is harmonic, $0 \notin f(D)$, and $1/f$ is also harmonic. Show that f or $\overline{f}$ is holomorphic.

Solution: Recall that f is harmonic if

$$\Delta f = 4\frac{\partial}{\partial z}\frac{\partial}{\partial \overline{z}} f = 0,$$

where

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right), \quad \frac{\partial}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right).$$

Computing $\Delta(1/f)$, we find that

$$0 = \Delta\left(\frac{1}{f}\right) = \frac{2}{f^3}\left(\frac{\partial f}{\partial z}\right)\left(\frac{\partial f}{\partial \overline{z}}\right) - \frac{1}{f^2}\Delta f = \frac{2}{f^3}\left(\frac{\partial f}{\partial z}\right)\left(\frac{\partial f}{\partial \overline{z}}\right).$$

Thus, given $z \in D$, either $\frac{\partial f}{\partial z} = 0$ or $\frac{\partial f}{\partial \overline{z}} = 0$. By the identity theorem, exactly one of these must hold for all $z \in D$. That is, by the C-R equations, either f is antiholomorphic (the first case) or holomorphic (the second case).

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Exercise 1.3 (September 2023, Problem 4): A complex function $w = W(z)$ can be expressed in polar coordinates: if $w = se^{i\gamma}$ and $z = re^{i\theta}$, then the two real functions of two real variables

$$s = S(r, \theta), \quad \gamma = \Gamma(r, \theta)$$

specify $w = W(z)$. Without changing to Cartesian coordinates, find necessary conditions on S and Γ for W to have a derivative W' at any point $z \neq 0$.

1.3 Elementary Functions

Let us make a quick detour to discuss how the exponential, logarithmic, and trigonometric functions generalize to the complex plane.

1.3.1 The complex logarithm

We already defined the complex exponential $e^z = e^x e^{iy}$. However, e^z is periodic with period $2\pi i$, so it is not entirely clear how to define the logarithm such that the logarithm is not multivalued. We define, for $z \neq 0$,

$$\log(z) = \log(|z|) + i\arg(z) = \log(|z|) + i(\mathrm{Arg}(z) + 2\pi k) \quad \forall k \in \mathbb{Z}.$$

Then it follows that $e^{\log(z)} = z$. However,

$$\log(e^z) = z + 2\pi k i \quad \forall k \in \mathbb{Z}.$$

Therefore, we define the *principal value of the logarithm* as

$$\mathrm{Log}(z) = \log(|z|) + i\mathrm{Arg}(z).$$

The function $\mathrm{Log}(z)$ is then well-defined and single-valued for $z \neq 0$. It is straightforward to verify that $\mathrm{Log}(z)$ is continuous on the domain

$$\mathcal{D} = \{z = re^{i\theta} \in \mathbb{C} : r > 0,\ -\pi < \theta < \pi\}.$$

Moreover, one can check that $\mathrm{Log}(z)$ satisfies the C-R equations on $\mathcal{D}$, so it is also holomorphic. The derivative of $\mathrm{Log}(z)$ is, as one might expect, $1/z$. More generally, $\log(z)$ is analytic on the domain

$$\mathcal{D}_\alpha = \{z = re^{i\theta} \in \mathbb{C} : r > 0,\ \alpha < \theta < \alpha + 2\pi\}.$$

The line $\theta = \alpha$ is called a *branch cut* of the logarithm, and this definition extends naturally to other multi-valued functions. An single-valued analytic function F that extends to a multi-valued function f is called a *branch* of f. The principal value of the logarithm is often called the principal branch of $\log(z)$.

It is now reasonable to define exponentiation of a complex number by another complex number, via $z^c = e^{c\log(z)}$.

Example 1.9: Express $(1+i)^{-i}$ in the form $x + iy$ for some real x, y.

Solution: We write

$$(1+i)^{-i} = e^{-i\log(1+i)} = e^{-i(\log\left(\sqrt{2}\right)+i(\pi/4+2\pi k))} = e^{\pi/4+2\pi k}\left(\cos\left(\log\sqrt{2}\right) - i\sin\left(\log\sqrt{2}\right)\right).$$

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Exercise 1.4 (September 2002, Problem 1): Find the real and imaginary parts of the complex number $(1+i)^i$.

Exercise 1.5 (September 2005, Problem 1):

(a) Determine all complex numbers z such that i^z has a finite numbers of values.

(b) Same question for i^{i^z}.

1.3.2 Trigonometric & hyperbolic functions

We define the complex sine and cosine functions by

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos(z) = \frac{e^{iz} + e^{-iz}}{2},$$

and the hyperbolic sine and cosine by

$$\sinh(z) = \frac{e^{z} - e^{-z}}{2}, \quad \cosh(z) = \frac{e^{z} + e^{-z}}{2}.$$

A direct consequence of these definitions are the identities

$$\sin(z) = -i\sinh(iz), \;\; \cos(z) = \cosh(iz), \;\; \sinh(z) = -i\sin(iz), \;\; \cosh(z) = \cos(iz).$$

1.4 Power Series

Recall that an analytic function on $\mathbb{R}$ is any function that can locally be defined as a convergent power series, and any such function is infinitely differentiable. We can extend this definition to complex functions.

A *power series* is a series expansion of the form

$$\sum_{k=0}^{\infty} a_k z^k$$

for some coefficients $a_k \in \mathbb{C}$.

Theorem 1.4 (Radius of Convergence)

For any power series $\sum_{k=0}^{\infty} a_k z^k$, there exists a *radius of convergence* $R \in [0, \infty]$ such that the series converges in the open disk $|z| < R$ and diverges in the open region $|z| > R$. Moreover, R can be found via the formulas:

$$1/R = \limsup_{n\to\infty} |a_n|^{1/n} \qquad \text{(Root Test)}$$

$$1/R = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| \qquad \text{(Ratio Test)}$$

One can show that every power series defined a holomorphic function on its radius of convergence. More generally, any analytic function on an open set is holomorphic on that set. The derivative of a power series is, of course, obtained by term-by-term differentiation. However, the proof is not entirely trivial. Note that holomorphic functions are also analytic, but the proof relies on integration results concerning holomorphic functions which we will cover in Day 2.

Example 1.10 (January 2016, Problem 2): Consider the power series

$$f(z) = \sum_{k=0}^{\infty} (-1)^k z^{2k}.$$

Determine the radius of convergence of $f(z)$. Does f have an analytic continuation to any region beyond this disk of convergence? Justify.

Solution: By the root test, we compute

$$1/R = \limsup_{n\to\infty} |a_k|^{1/k} = 1,$$

so $f(z)$ converges in the disk $|z| < 1$. Observe now that

$$f(z) = \sum_{k=0}^{\infty} (-1)^k z^{2k} = \sum_{k=0}^{\infty} (-z^2)^k = \frac{1}{1+z^2}.$$

The function $g(z) = 1/(1+z^2)$, defined on $\mathbb{C}\backslash\{i, -i\}$ then agrees with $f(z)$ on the disk $|z| < 1$, so by analytic continuation (the identity theorem), f can be extended to $g(z)$.

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Exercise 1.6 (September 2003, Problem 2): By analytic continuation beyond the disk $|z+1| < 1$, determine $\lim_{z\to 1} f(z)$, where $f(z)$ is the analytic function defined in the disk $|z+1| < 1$ by

$$f(z) = \sum_{n=0}^{\infty} (n+1)(n+2)(z+1)^n.$$

Day 2: Complex Integration

2.1 Contour Integrals

Of particular importance in complex analysis are integrals over closed curves. Before reviewing the key theorems about the integral of a holomorphic function over a closed curve, let us review contour integration.

We define a *curve* C in $\mathbb{C}$ as a continuous function $\gamma\colon [0,1] \to \mathbb{C}$ with $\gamma(t) = x(t)+iy(t)$. The point $\gamma(0)$ is called the *initial point* and $\gamma(1)$ the *terminal point*. If the initial and terminal points are the same, the curve is called *closed*. Note that we could replace $[0,1]$ in the definition with any real closed interval $[a,b]$ with $a < b$. The curve γ is called *differentiable* if the component functions x and y are continuously differentiable functions of t, with

$$\gamma'(t) = x'(t) + iy'(t),$$

and such that $\gamma'(t) \neq 0$ for all $t \in [0,1]$. A *contour* is any piecewise differentiable curve. We are mostly interested in *simple curves*, that is, curves that do not intersect themselves. We use the term *positively oriented* to describe a curve that is oriented in the counterclockwise direction.

Given a contour C parametrized by γ on $[a,b]$, we define

$$\int_C f = \int_C f(z)\ dz := \int_a^b f(\gamma(t))\gamma'(t)\ dt.$$

Exercise 2.1: Compute the integral $I = \int_C \overline{z}\ dz$ when C is given by the curve $z = 2e^{i\theta}$, where $\theta \in [-\pi/2, \pi/2]$.

Example 2.1: Compute the integral $I = \int_C \sqrt{z}\ dz$ where C is the quarter-circle arc from $z = 1$ to $z = i$ and the integrand is taken with the principal branch.

Solution: The contour C here is parametrized by the curve $\gamma(t) = e^{it}$ for $t \in [0, \pi/2]$. We have

$$I = \int_0^{\pi/2} e^{it/2}(ie^{it})\ dt = i\int_0^{\pi/2} e^{i(3t/2)}\ dt = \frac{2}{3}\left(-\frac{1}{\sqrt{2}} - 1 + i\frac{1}{\sqrt{2}}\right).$$

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Exercise 2.2: Redo Example 2.1 where the branch is given by $\pi < \theta < 3\pi$. Why is your answer different?

2.2 Analyticity and Integration

In this section, we will demonstrate how analyticity is a much stronger property than mere differentiability. It is likely that a significant portion of the written exam will require the use of the main results of this section, either explicitly or implicitly.

2.2.1 Antiderivatives & Cauchy's Theorem

We begin with the antiderivative theorem:

Theorem 2.1 (Antiderivative Theorem)

Let f be continuous on a domain D. Then the following are equivalent:

(i) $f(z)$ has an antiderivative $F(z)$ in D;

(ii) The contour integral of f from z_1 to z_2 in D is independent of path. More specifically,

$$\int_\Gamma f = F(z_2) - F(z_1)$$

for any contour Γ from z_1 to z_2;

(iii) For any closed contour C in D,

$$\int_C f = 0.$$

Example 2.2: Evaluate the integral $\int_\gamma ze^z\, dz$, where γ is the path parametrized by $\gamma(t) = t + i\sin^2(\pi t)$ for $t \in [0,1]$.

Solution: The integrand as antiderivative $(z-1)e^z$, so we simply evaluate

$$\int_\gamma ze^z\, dz = (z-1)e^z\big|_{z=0}^{z=1} = 1.$$

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Exercise 2.3: Let $g(z) = 1/z$, defined on $D = \mathbb{C}\backslash\{0\}$. Show that g does not have an antiderivative in D.

Perhaps the most important theorem in complex analysis is Cauchy's theorem (also known as Goursat's theorem, or Cauchy-Goursat):

Theorem 2.2 (Cauchy-Goursat)

Let $D \subseteq \mathbb{C}$ be a domain whose boundary is given by a simple contour $C = \partial D$. Suppose $f \in \mathcal{H}(D) \cap C(\overline{D})$. Then

$$\int_{\partial D} f = 0.$$

In particular, for any simple contour $C \subseteq D$, the integral of f over C vanishes. Combined with the antiderivative theorem, this shows that every holomorphic function on D admits an antiderivative on D.

An immediate corollary of Cauchy's theorem is the "holes theorem," which allows one to reduce the integral of f around C to the sum of integrals around smaller contours C_k which contain the "holes" of the domain D. More precisely, if C_k are positively-oriented simple closed contours in D such that f is analytic in $D \setminus \bigcup_{k=1}^{N} C_k$, then

$$\int_{\partial D} f = \sum_{k=1}^{N} \int_{C_k} f.$$

2.2.2 Cauchy's Integral Formula

The first consequence of Cauchy's theorem is the Cauchy Integral Formula (CIF).

Theorem 2.3 (Cauchy Integral Formula)

Let D be a domain bounded by a simple closed contour ∂D. Suppose $f \in \mathcal{H}(D) \cap C(\overline{D})$. Then for every $z_0 \in D$,

$$f(z_0) = \frac{1}{2\pi i} \int_{\partial D} \frac{f(z)}{z - z_0}\, dz.$$

The proof uses that $\int_{\partial D} dz/(z - z_0) = 2\pi i$ and considers the difference quotient of f at z_0. The following generalization is also useful:

Theorem 2.4 (Cauchy Derivative Formula)

Under the same hypotheses as above,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{\partial D} \frac{f(z)}{(z - z_0)^{n+1}}\, dz.$$

A consequence of the CIF is that all derivatives of analytic functions are also analytic. We also have the following simple estimate:

Theorem 2.5 (Cauchy's Inequality)

Suppose f is holomorphic on $B(z_0, R)$ and continuous on $\overline{B(z_0, R)}$. Then

$$\left|f^{(n)}(z_0)\right| \leq \frac{n! M_R}{R^n},$$

where $M_R = \sup_{\partial B(z_0,R)} |f(z)|$.

Exercise 2.4 (Liouville's Theorem): Any entire and bounded function is constant.

Example 2.3 (September 2024, Problem 4): Classify all entire functions $f(z)$ that satisfy $|f(z)| \leq \sqrt{1 + |z|}$ for all $z \in \mathbb{C}$.

Solution: Let $z_0 \in \mathbb{C}$ be given and take any circle C_R centered at z_0. Let $z \in C_R$. Then

$$|z| \leq |z_0| + R,$$

and

$$|f(z)| \leq \sqrt{1 + R + |z_0|} \leq 1 + \sqrt{R} + \sqrt{|z_0|}.$$

Thus, by Cauchy's inequality

$$|f'(z_0)| \leq \frac{1}{R}\left(1 + \sqrt{R} + \sqrt{|z_0|}\right) \to 0 \text{ as } R \to \infty.$$

Thus, $f \equiv a$ for some $a \in \mathbb{C}$. Moreover,

$$|a| \leq \inf_{z \in \mathbb{C}} \sqrt{1 + |z|} = 1,$$

so any such f is a constant within $\overline{B(0,1)} \subseteq \mathbb{C}$.

■

2.2.3 Morera's Theorem and Maximum Modulus Principle

Here we present two important consequences of Cauchy's theorem that may show up on the written exam. The first is combines the antiderivative theorem with the fact that the derivatives of holomorphic functions are holomorphic:

Theorem 2.6 (Morera)

Let D be a domain. Suppose $f \in C(\overline{D})$ satisfies $\int_C f = 0$ for every simple closed contour C in D. Then f is analytic in D.

Example 2.4: Let D be a domain and $\{f_n\}_{n=1}^{\infty}$ be a sequence of holomorphic functions on D. Suppose $f_n \to f$ uniformly on every compact subset of D. Show that f is holomorphic on D.

Solution: Let $\overline{B} \subseteq D$ be any closed ball in D, and $\Delta \subseteq \overline{B}$ any triangle. Since each f_n is holomorphic in D, we have $\int_\Delta f_n = 0$. Moreover, the convergence $f_n \to f$ is uniform, so f is continuous, and

$$\int_\Delta f = \lim_{n \to \infty} \int f_n = 0.$$

By Morera's theorem, f is analytic in $\overline{B}$, hence in D.

■

Exercise 2.5 (January 2013 Problem 1(b)): For which values of $z \in \mathbb{C}$ does the series

$$f(z) = \sum_{n=1}^{\infty} \frac{\cos(nz)}{e^n}$$

converge. For which values of z is $f(z)$ analytic? *Hint: Use the previous example.*

It is also often convenient to rewrite the CIF in *Gauss mean value theorem* form: let $z_0 \in \mathbb{C}$ and let C_ρ be a small positively-oriented circle $|z - z_0| = \rho$ around z_0. Then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta}) d\theta.$$

That is, for holomorphic f, it follows that $f(z_0)$ is the arithmetic mean of its values on C_ρ. This can be used to show that if $|f(z)| \leq |f(z_0)|$ in a neighbourhood of z_0, then f is constant on this neighbourhood with value $f(z_0)$. The maximum modulus principle now follows:

Theorem 2.7 (Maximum Modulus Principle)

If f is a non-constant holomorphic function on a domain D, then $|f(z)|$ does not attain a maximum in D.

A simple corollary of this is that if D is bounded and f is continuous on $\overline{D}$, then the extrema of $|f(z)|$ must lie on ∂D.

Example 2.5 (September 2022, Problem 4): Let f be holomorphic on some open set containing the closed unit disk $|z| \leq 1$. Assume that

$$|1 - f(z)| \leq \left| e^{z-1} \right|$$

on $|z| = 1$. Prove that $1/2 \leq |f(0)| \leq 3/2$.

Solution: We have, for $|z| = 1$,

$$\left| \frac{1 - f(z)}{e^z} \right| \leq \frac{1}{e}.$$

Since f is holomorphic in the disk, the maximum modulus principle applies to the above. In particular, for $z = 0$ we have

$$|1 - f(0)| \leq \frac{1}{e} \leq \frac{1}{2},$$

from which the desired bounds are derived.

■

Example 2.6 (September 2017, Problem 5): Let f_k for $k = 1, \ldots, n$ be holomorphic in a domain D. Can the function

$$f(z) = \sum_{k=1}^{n} |f_k(z)|$$

have a strict local maximum in D? What about a strict local minimum?

Solution: We claim that f cannot have a strict local maximum in D. Assume instead that z_0 is a maximizer of f. Let C_ρ be a small circle around z_0 with radius ρ such that $f(z) \leq f(z_0)$ for all $z \in C_\rho$. Then

$$f_k(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f_k(z_0 + \rho e^{it})dt$$

for each $k = 1, \ldots, n$. Thus,

$$\begin{aligned} f(z_0) = \sum_{k=1}^{n} |f_k(z_0)| \leq \sum_{k=1}^{n} \frac{1}{2\pi}\int_0^{2\pi} \left|f_k(z_0 + \rho e^{it})\right| \, dt &= \frac{1}{2\pi}\int_0^{2\pi} \left(\sum_{k=1}^{n} \left|f_k(z_0 + \rho e^{it})\right|\right) dt \\ = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + \rho e^{it}) \, dt \leq f(z) < f(z_0), \end{aligned}$$

a contradiction. Therefore, f cannot have a strict local minimum in D.

A strict local minimum is fine. For instance, let $n = 1$, $f_1(z) = z$ and $f(z) = |z|$.

■

2.2.4 Analytic Continuation

Let us make precise the notion of analytic continuation, which we have used to solve some of the previous examples. The first result we require is the identity theorem.

Theorem 2.8 (Identity Theorem)

Let h be holomorphic on a domain D. Then the following are equivalent:

(i) $h \equiv 0$ in D;

(ii) The set $D_0 = \{z \in \mathbb{C} : h(z) = 0\}$ has a limit point.

Proof. Choose $w_k \in D_0$ such that $w_k \to z_0$. We first show that f vanishes in a disk $B \subseteq D$ around z_0. Consider the power series

$$f(z) = \sum_{k \geq 0} a_k (z - z_0)^k.$$

If f were not identically zero in B, then we can choose the smallest integer N such that $a_N \neq 0$. Then

$$f(z) = a_N(z - z_0)^N(1 + g(z - z_0))$$

for some analytic function g that vanishes as $z \to z_0$. Plugging the w_k into this expression then provides a contradiction.

To conclude that f vanishes on all of D, let U denote the interior of the points where f vanishes. We just showed that U is non-empty, and, by definition, U is open. But U is also closed, since f is continuous. Now, the domain D is connected, so in fact $U = D$.

$\square$

Now, if f and g are two analytic functions, the identity theorem can be readily applied to $h = f - g$ to show that $f = g$ on a domain D. If f is defined on an open set $U \subseteq \mathbb{C}$, and $V \subseteq \mathbb{C}$ is another open set such that $U \subseteq V$. Then if F is an analytic function on V such that $F|_U = f$, we call F an *analytic continuation* of f.

2.3 More Exercises

Exercise 2.6 (January 2025, Problem 1): Let $C_2 = \{z \in \mathbb{C} : |z| = 2\}$. Use **CIF** to compute

$$\int_{C_2} \frac{1}{z^4 + 1} dz.$$

Exercise 2.7 (September 2023, Problem 1): State whether the following statements are true or false (a one or two-line justification for each will suffice).

(a) There exists a function which is analytic on the unit disk and such that

$$\int_{|z|=1/2} f = -1.$$

(b) There exists an unbounded entire function f such that $\mathrm{Re}(f) \equiv -1$.

(c) The power series $\sum_{n\geq 0} n^n z^n$ converges at some complex number $z \neq 0$.

(d) There exists a nonzero function f which is analytic on the unit disk, and such that $f^{(n)}(0) = 0$ for all $n \geq 0$.

(e) Assume f is an entire function such that $|f(0)| \geq |f(z)|$ for all $z \in \mathbb{C}$, and let $f(z) = \sum_{n\geq 0} a_n z^n$ be its Taylor series. Then $a_5 = 0$.

Exercise 2.8 (January 2011, Problem 3): The function f is analytic in the whole plane and have positive imaginary part. What can it be? What if all you know is that the imaginary part of f tends to 0 at ∞?

Day 3: Laurent Series & Residues

3.1 Laurent Series

Any function f which is holomorphic at a point z_0 admits a Taylor series at z_0. This fact is yet another consequence of the CIF. Thus, holomorphic and analytic are synonymous. A *Laurent series* generalizes the Taylor series to functions that are analytic except for at a point.

Theorem 3.1

Suppose f is analytic throughout some annulus $R_1 < |z - z_0| < R_2$ centered at z_0, and let C be any simple closed contour around z_0 contained in this domain. Then f admits the series representation

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} b_n(z - z_0)^{-n}$$

in $R_1 < |z - z_0| < R_2$. The coefficients are given explicitly by

$$a_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz, \quad b_n = \frac{1}{2\pi i}\int_C \frac{f(z)}{(z - z_0)^{1-n}}\, dz.$$

The proof, like for Taylor's theorem, relies on some CIF shenanigans. Computing the Laurent series in a given annulus is a basic exercise that often shows up on exams.

Example 3.1 (September 2025, Problem 3): Consider the complex function

$$f(z) = \frac{1}{z(z-1)(z-2)}.$$

Obtain the Laurent series expansion in the annulus $1 < |z| < 2$.

Solution: First, let us split up f via partial fractions to obtain

$$f(z) = \frac{1}{2z} + \frac{1}{1-z} - \frac{1}{2(2-z)}.$$

The first term is already of the form $1/z$, so we can leave it alone for now. Since $|z| > 1$, we rewrite the second term as

$$\frac{1}{1-z} = -\frac{1}{z} \cdot \frac{1}{1-(1/z)} = -\frac{1}{z}\sum_{n=0}^{\infty} \frac{1}{z^n} = -\sum_{n=1}^{\infty} \frac{1}{z^n}.$$

The last term becomes

$$\frac{1}{2(2-z)} = \frac{1}{4(1-(z/2))} = \sum_{n=0}^{\infty} \frac{z^n}{2^{n+2}}.$$

Overall, the Laurent series is

$$f(z) = \left(-\frac{1}{2}\right)\frac{1}{z} - \sum_{n=2}^{\infty} \frac{1}{z^n} - \sum_{n=0}^{\infty} \frac{z^n}{2^{n+2}}.$$

■

Exercise 3.1: Find the Laurent series expansion of $f(z)$ from Example 3.1 in the regions $0 < |z| < 1$ and $|z| > 2$.

3.2 Singularities of Analytic Functions

3.2.1 Cauchy's Residue Theorem

If f is analytic on a domain D except at finitely many points $z_1, \ldots, z_n$, we can define the *residue* of f at the point z_k by

$$\operatorname{Res}_{z=z_k} f(z) \coloneqq 2\pi i b_1^{(k)},$$

where $b_1^{(k)} = \int_{C_k} f$ is the first coefficient of the Laurent series expansion of f at the point z_k. Here, C_k is a positively oriented simple closed contour contained in D that contains z_k. Thus,

$$\int_{C_k} f = 2\pi i \operatorname{Res}_{z=z_k} f(z).$$

Theorem 3.2 (Cauchy's Residue Theorem)

Let D be a domain bounded by a (positively oriented) simple closed curve ∂D. Suppose f is analytic in D except at finitely many points $z_1, \ldots, z_n \in D$, and that f is continuous on $\overline{D}$. Then

$$\int_{\partial D} f = 2\pi i \sum_{k=1}^{n} \operatorname{Res}_{z=z_k} f(z).$$

This is, in effect, exactly the same thing as the previously discussed "holes theorem" (see Theorem 2.2).

Exercise 3.2 (September 2023, Problem 3): Consider the function

$$f(z) = \frac{\sin(z)}{z^3(z-1)^2}.$$

(a) Find the first two non-zero terms of the Laurent series of f in the annulus $0 < |z| < 1/2$, and compute $\int_{C_0} f$ where C_0 is the circle of radius $1/4$ centered at $z = 0$, oriented CCW.

(b) Repeat part (a) for the annulus $0 < |z-1| < 1/2$ and C_1 the circle of radius $1/4$

centered at $z = 1$, oriented CCW.

3.2.2 Poles and zeroes of analytic functions

If f is analytic except at a point z_0, the part of the Laurent series of f with negative powers of $(z - z_0)$,

$$\frac{b_1}{(z-z_0)} + \frac{b_2}{(z-z_0)^2} + \ldots + \frac{b_k}{(z-z_0)^k} + \ldots$$

is called the *principal part*. If the principal part terminates at the power $k = N$, then the point z_0 is called a *pole of order N of f*. In the case $N = 1$, z_0 is often called a *simple pole*. If each $b_k = 0$, then z_0 is called a *removable singularity*, and if the b_k never terminate, z_0 is an *essential singularity*. Essential singularities are particularly strange (we will state a few theorems concerning them later).

A convenient characterization of poles that is often used when computing residues is the following:

Theorem 3.3

Let f be analytic except at an isolated singular point z_0. Then z_0 is a pole of order N if and only if

$$f(z) = \frac{\phi(z)}{(z-z_0)^N}$$

for some analytic function ϕ with $\phi(z_0) \neq 0$. Moreover, if $N = 1$,

$$\mathrm{Res}_{z=z_0} f(z) = \phi(z_0),$$

and if $N > 1$,

$$\mathrm{Res}_{z=z_0} f(z) = \frac{\phi^{(N-1)}(z_0)}{(N-1)!}.$$

It is straightforward to define ϕ given the Laurent series of f by setting

$$\phi(z) = \begin{cases} (z-z_0)^N f(z) & z \neq z_0 \\ b_N & z = z_0 \end{cases}.$$

Exercise 3.3 (January 2025, Problem 1 (again)): Let $C_2 = \{z \in \mathbb{C} : |z| = 2\}$. Use **residues** to compute

$$\int_{C_2} \frac{1}{z^4+1} dz.$$

Let us also quickly discuss the zeroes of analytic functions so that we can state another nice formula for computing contour integrals. If $f(z_0) = 0$ and N is the smallest integer such that $f^{(N)}(z_0) \neq 0$, then z_0 is called a *zero of order N of f*.

Theorem 3.4

Let f is analytic at z_0. Then z_0 is a zero of order N of f if and only if there is an analytic function g with $g(z_0) \neq 0$ such that $f(z) = (z - z_0)^N g(z)$.

A nice corollary of this is that the zeros of analytic functions are isolated (Exercise: prove this). From this, it is easy to see that if p, q are analytic functions such that $p(z_0) \neq 0$ and q has a zero of order N at z_0, then p/q has a pole of order N at z_0. Now, for simple poles, this provides a simple formula for finding residues:

Theorem 3.5

Let p, q be analytic at z_0 such that $p(z_0) \neq 0$ and q has a zero of order 1 at z_0. Then

$$\operatorname{Res}_{z=z_0} \frac{p(z)}{q(z)} = \frac{p(z_0)}{q'(z_0)}.$$

Example 3.2 (January 2012, Problem 2): Compute

$$\int_C \frac{z \sec(z)}{(1 - e^z)^2} \, dz,$$

where C is the circle of radius 2 centered at the origin.

Solution: We first rewrite the integrand as

$$f(z) = \frac{z \sec(z)}{(1 - e^z)^2} = \frac{z}{\cos(z)(1 - e^z)^2},$$

from which we can identify $z = 0, \pm\pi/2$ as simple poles. We compute the residues at each pole:

For $\pi/2$, we compute

$$\operatorname{Res}_{z=\pi/2} f(z) = \frac{\pi/2}{(-\sin(\pi/2)(1 - e^{\pi/2})^2 - 2\cos(\pi/2)e^{\pi/2}(1 - e^{\pi/2}))} = -\frac{\pi}{2(1 - e^{\pi/2})^2}.$$

Likewise,

$$\operatorname{Res}_{z=-\pi/2} f(z) = -\frac{\pi}{2(1 - e^{-\pi/2})^2}.$$

For $z = 0$, we first observe that

$$\left(\frac{(1 - e^z)^2}{z}\right)' = \frac{-2ze^z(1 - e^z) - (1 - e^z)^2}{z^2}.$$

Thus, after two applications of L'Hopital, we find that

$$\operatorname{Res}_{z=0} f(z) = \lim_{z \to 0} \frac{z^2}{-2ze^z(1 - e^z) - (1 - e^z)^2} = 1.$$

Overall,
$$\int_C \frac{z\sec(z)}{(1-e^z)^2}\ dz = 2\pi i\left(1 - \frac{\pi}{2(1-e^{\pi/2})^2} - \frac{\pi}{2(1-e^{-\pi/2})^2}\right).$$

■

3.2.3 Computing real integrals using residues

Computing a real integral via complex-analytic methods is possibly the most likely question to appear on the written exam. We will cover two examples, and a few more important exercises are left below those.

Example 3.3 (September 2024, Problem 1): Compute the following integral using the residue theorem:
$$\int_{-\infty}^{\infty} \frac{x}{(x^2+2x+2)^2}\ dx.$$

Solution: We consider the complexified (is this a word people use?) version of the integrand
$$f(z) = \frac{z}{(z^2+2z+2)^2},$$
and observe that f has two poles, each of order 2, at
$$z_0 = -1+i, \quad z_1 = -1-i.$$
Let C_R be the positively oriented semicircle of radius R in the closed upper half-plane, $R \gg 1$. Moreover, let Γ_R be the interval $[-R, R]$ on the real axis, and define $C = \Gamma_R \cup C_R$, oriented positively. We compute the residue of f as z_0 as follows:
$$\begin{aligned}\operatorname{Res}_{z=z_0} f(z) = \int_C \frac{z/(z+1+i)^2}{(z-(-1+i))^2}\ dz &= 2\pi i\left(\frac{z}{(z+1+i)^2}\right)'\Big|_{z=-1+i} \\ &= 2\pi i\frac{(2i)^2 - 2(-1+i)(2i)}{(2i)^4} \\ &= -\frac{\pi}{2},\end{aligned}$$
by CIF. The next step is to show that the part on the arc of the circle vanishes as we take R to infinity. To this end, fix $z \in \mathbb{C}$ with $|z| = R$. Then
$$|f(z)| \leq \frac{R}{(R^2-2R-2)^2},$$
where we have applied the triangle inequality. Thus,
$$\lim_{R\to\infty}\left|\int_{C_R} f(z)\ dz\right| \leq \lim_{R\to\infty} \pi\frac{R^2}{(R^2-2R-2)^2} = 0.$$
Therefore,
$$\int_{-\infty}^{\infty} \frac{x}{(x^2+2x+2)^2}\ dx = \int_C \frac{z}{(z^2+2z+2)^2}\ dz = -\frac{\pi}{2}.$$

■

Example 3.4 (January 2007, Problem 1): Show that $\sum_{n\geq 1} 1/n^2 = \pi^2/6$ by integrating $1/(z^2(e^{2\pi iz}-1)$ along the boundary of a suitable box.

Solution: First note that
$$f(z) = \frac{1}{z^2(e^{2\pi iz}-1)}$$
has a triple pole at z and simple poles at $z=k$ for every integer $k\neq 0$. It is a straightforward computation to show that
$$\operatorname{Res}_{z=k} f(z) = \frac{1}{2\pi ik^2}.$$
We must choose a contour that avoids all the roots. To this end, consider the box

(I)
$$z = x - i\left(N+\frac{1}{2}\right),$$

(II)
$$z = \left(N+\frac{1}{2}\right) + iy,$$

(III)
$$z = x + i\left(N+\frac{1}{2}\right),$$

(IV)
$$z = -\left(N+\frac{1}{2}\right) + iy,$$

where
$$x, y \in \left[-\left(N+\frac{1}{2}\right), N+\frac{1}{2}\right]$$
and the box C is to be oriented positively.

On (I) and (III), observe that
$$\lim_{N\to\infty}\left|\int_\Gamma f(z)\ dz\right| \leq \lim_{N\to\infty} 2\left(N+\frac{1}{2}\right)\frac{1}{(N+1/2)^2(e^{-2\pi(N+1/2)}-1)} \to 0,$$
by the triangle inequality. On the contours (II) and (IV), we find
$$e^{2\pi iz} = e^{2\pi iN}e^{\pi i}e^{-2\pi y} = -e^{-2\pi y}.$$
Then the contour integrals over (II) and (IV) again decay like $1/N$. By Cauchy's theorem
$$2\sum_{n=1}^{\infty}\frac{1}{n^2} + 2\pi i\operatorname{Res}_{z=0} f(z) = 0.$$

To compute $\operatorname{Res}_{z=0} f(z)$, one can do series division to find

$$\frac{1}{z^2(e^{2\pi iz}-1)} = \frac{1}{2\pi iz^3} \cdot \frac{1}{1+\pi iz+\frac{(2\pi i)^2}{6}z^2+\dots} = \frac{1}{(2\pi i)z^3} - \frac{1}{2z^2} + \frac{i\pi}{6} \cdot \frac{1}{z} + \dots,$$

from which we identify $\operatorname{Res}_{z=0} f(z) = (\pi i)/6$.

■

Exercise 3.4 (September 2012, Problem 3): Show that

$$\int_0^\infty \frac{1}{1+x^3}\,dx = \frac{2\pi}{3\sqrt{3}}$$

by integrating around the contour which consists of arc of the circle of radius R for $0 \le \theta \le 2\pi/3$, where it becomes the line segment $re^{i2\pi/3}$ for $0 \le r \le R$.

Exercise 3.5 (September 2016, Problem 3): For any real number $p > 1$, calculate

$$\int_0^\infty \frac{1}{x^p+1}\,dx.$$

Exercise 3.6 (September 2007, Problem 2): Calculate the integral

$$I = \int_0^\infty \frac{\cos(x)}{x^2+1}\,dx.$$

Exercise 3.7 (January 2011, Problem 1): Calculate the integral

$$I = \int_{-\infty}^\infty \frac{1}{x^4+3x^2+4}\,dx.$$

Exercise 3.8 (January 2017, Problem 5(b)): Let n be an arbitrary positive integer. Evaluate

$$\int_0^\infty \frac{1}{x^{2n}+x^{-2n}}\,dx.$$

Hint: Use a suitable shaped and positioned "pizza slice" contour.

3.3 Consequences of the Residue Theorem

3.3.1 Some facts concerning essential singularities

3.3.2 Arugment principle & Rouche's theorem

Day 4: Conformal Mappings

4.1 Linear Fractional Transformations

4.2 The Schwarz Lemma

4.2.1 The lemma

4.2.2 Automorphisms of the disk and upper half-plane

4.3 Riemann Mapping Theorem